DA11 Charts that say what the numbers say

Week 4, Clean and chart | Session 11 of 24 | Python for Data Analysis | New library: matplotlib

Today's question. Which chart shows this comparison most clearly?

Objective. By the end: you can draw a bar chart, a line chart, a histogram and a scatter plot with labelled axes, and state one finding and one limit for each.

Before this session: DA10 Cleaning without hiding evidence. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
matplotlib	The library that draws charts. We load it as plt.	A pen and a ruler for drawing.
bar chart	Bars compare amounts between categories.	Piles of stock side by side.
line chart	A line shows how one thing changes over ordered dates.	A temperature record.
histogram	Bars show how many values fall in each range.	Counting pupils in each height band.
scatter plot	Dots show how two numbers go together.	Each pupil as a dot by height and age.
axis label	The words along an axis that name the quantity and its unit.	The heading of a column.

Block A. Bar and line charts

plt.bar draws one bar for each category. A bar chart starts its value axis at zero, because the length of a bar stands for the amount. plt.plot joins points in order, which suits dates.

Predict. Predict which of the three bars will be tallest. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
import matplotlib.pyplot as plt
shop = pd.read_csv("camp_shop.csv")
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
by_item = shop.groupby("item")["revenue_ugx"].sum()
plt.bar(by_item.index, by_item.values)
plt.ylabel("Revenue (UGX)")
plt.title("Revenue by item, practice shop")
plt.ylim(bottom=0)
plt.show()
by_date = shop.groupby("date")["revenue_ugx"].sum()
plt.plot(by_date.index, by_date.values)
plt.ylabel("Revenue (UGX)")
plt.xticks(rotation=90)
plt.show()
```

Run it, then compare. The first chart has three bars. Book is tallest at 297000, then Juice and Pen. The second chart is a line of 20 daily revenues, with the highest point on 2026-12-20. Every axis says what it measures and the bar chart starts at zero.

Investigate.

1. Why does the bar chart start at zero?
2. Why use a line for dates and bars for items?

Modify. Draw the bar chart of units by item and add its label.

Block B. Histogram and scatter plot

A histogram shows how the values are spread. A scatter plot puts two numbers on the two axes. These two charts use the Sunrise Shops tables.

Predict. Predict the shape of the histogram of sale values: balanced, or bunched at the low end with a long tail? Write your guess on paper first.

Your guess, written before you run it

```
sunrise = pd.read_csv("sunrise_sales.csv")
sunrise["line_value"] = sunrise["units"] * sunrise["unit_price_ugx"]
plt.hist(sunrise["line_value"], bins=30)
plt.xlabel("Value of one sale (UGX)")
plt.ylabel("Number of sales")
plt.show()
daily = pd.read_csv("sunrise_daily_summary.csv")
plt.scatter(daily["customers_counted"], daily["revenue_ugx"])
plt.xlabel("Customers counted in a day")
plt.ylabel("Revenue that day (UGX)")
plt.show()
```

Run it, then compare. The histogram is bunched at the low end with a long tail to the right, because most sales are small and a few are very large. The scatter plot has 180 dots, one for each shop and day. More customers usually mean more revenue, but the dots are scattered, so the link is not exact.

Investigate.

1. What does the long tail in the histogram tell us?
2. Does the scatter plot show that customers cause revenue?

Modify. Change bins=30 to bins=10 and describe what the histogram loses.

Make

Chart the Sunrise revenue by branch. Use a dictionary to turn the ids 1, 2 and 3 into the names Central Market, Campus Gate and Roadside. Label the axis, start it at zero and write one finding and one limitation.

Save your work as DA11. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
The chart appears empty	plt.show() ran before the data was added, or the data was empty.	Run the cell that creates the table first and print its length.
The labels on the dates overlap	There are too many labels to fit side by side.	Rotate them with plt.xticks(rotation=90) or plot fewer dates.

Exit ticket

I choose the chart by the question, label its axes, and say what it cannot show.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.